# Hassan Elseoudy

Senior Software Engineer | Berlin, Germany

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## Summary

Senior software engineer with 8+ years designing distributed backend systems in Java, Kotlin, and Python. At Zalando, designs and leads cross-team platforms for supplier finance and logistics, and drives platform modernization and AI-assisted engineering. Internal speaker, Agentic AI Labs trainer, and startup founder.

## Professional Experience

**Senior Software Engineer** **Zalando** Berlin, Germany 09/2022 – Present

- **Designed the architecture and led delivery** of a **supplier-chargeback platform** that automatically bills suppliers for contract violations; its reusable design scaled to **seven chargeback types** generating **€33M+ a year** (€67M+ since 2025) across **1,200+ suppliers** and saved ~**1.5 FTEs** of manual work.
- Authored the technical designs for a **supplier self-service tool** spanning **six services** and for clarification workflows covering ~**500 suppliers**; the launch saved or avoided ~**3.5 FTEs** and cut delivery delays by **1.5 days**.
- Developed and operated the system suppliers use to announce deliveries (**1.5M+ notices**, **250M+ row** database, **99.99% availability**) and its **transport-planning** integration for scheduling and warehouse capacity.
- Automated **customs import filings** and built the tariff and duty-rate services used to price imported goods.
- Delivered core parts of the **About You integration**: made purchasing rules configurable instead of hard-coded, and built the system for internal sales and stock-ownership transfers between the two companies.
- Led the **Spring Boot 3 and 4 migrations** across **10+ services**, upgrading internal shared packages and external dependencies; extended **shared messaging libraries used by other teams** with tracing and event deduplication.
- Built **Shipping Notice Copilot**, an AI prototype that pre-fills supplier delivery notices from order history; the team won **"Best AI Usage"** at Procure-to-Pay HackWeek 2026.
- Selected as an **Agentic AI Labs trainer** for AI agents and MCP, and designing a hands-on workshop for ~**20 engineers across three teams**.
- Represented the shipment domain in **cross-team architecture** discussions; onboarded **three engineers**, interviewed candidates, and led incident root-cause analyses.

*Tech:* Java, Kotlin, Spring Boot, PostgreSQL, Kafka (Nakadi), Kubernetes, AWS, OPA/Rego, OpenTelemetry

**Founder & CEO** **Noaat** (Restaurant & retail tech startup) Alexandria, Egypt 01/2024 – 01/2026

- **Founded and scaled a B2B SaaS platform from 0→1**, serving **50+ restaurant clients** at **99.9% uptime**, and evolved it from a **customer feedback platform** (surveys with response analytics) and loyalty app into a cashback marketplace and a point-of-sale (POS) system.
- **Built and led a cross-functional team of 8 engineers** (frontend, mobile, backend, QA) plus data and design; set the technical roadmap, ran **40 two-week sprints**, defined code-review standards and on-call playbooks, and was the main reviewer (**330+ reviews**).
- **Architected the full technology stack**: a **multi-tenant** Kotlin/Spring Boot backend (~**240 API endpoints**) supporting **50+ branches** with role-based access control, staff permissions, and centralized client management; a React Native app, a Next.js website, and QR table ordering on AWS.
- Built an **enterprise point-of-sale system** with real-time order management, kitchen display integration, and thermal printing through a custom Android printer bridge; personally built the POS web app (40 releases).
- Implemented a **coupon and cashback loyalty system** with receipt-based claims and automatic receipt verification (AWS Textract, OpenAI), driving a **30% increase** in repeat customer visits.
- Built an **Arabic AI sales and support chatbot** on Facebook Messenger (n8n, GPT-4o-mini, Pinecone) that answers customers, opens Jira tickets for issues, and routes sales leads to the CRM and Slack.
- Led a cloud security and reliability overhaul: moved the AWS setup to version-controlled infrastructure, ran incident response, blocked an SMS-verification flood attack, and scaled the backend horizontally.
- **Drove product-market fit** through direct client engagement, iterating on features and closing enterprise accounts.

*Tech:* Kotlin, Spring Boot, PostgreSQL, Redis, TypeScript, React, React Native, Next.js, AWS (ECS, RDS, Lambda, S3, CloudFormation, Textract), OpenAI, n8n

**Senior Backend Engineer (Part-time)** **Cynopsis Solutions** Singapore 10/2021 – 03/2026

- Scaled AML/KYC transaction processing to **1M+ daily transactions**, improving throughput by **300%**.
- Designed and implemented **20+ configurable compliance rules** covering transaction velocity, structuring detection, cross-border flows, and watchlist screening.
- Led the AI integration strategy, building a **RAG-powered compliance assistant** that cut analyst investigation time by **60%** and enabled natural-language regulatory queries.
- Directed **platform modernization**: multi-tenant architecture, CRM-based user provisioning, audit-ready permissions, and Arabic right-to-left support enabling expansion into **Middle East markets**.
- Designed a **case-workflow engine** with configurable status transitions, escalation paths, and ageing calculations for **regulatory SLA compliance**.
- Built an **executive Metabase dashboard** (case distribution, risk bucketing, analyst performance) and authored **API and technical design documents** that sped up onboarding and integrations.

*Tech:* Python, Django, LangChain, Celery, TimescaleDB, PostgreSQL, vector databases, Metabase

**Java Engineer** **DHL Freight Enterprise** Sofia, Bulgaria 10/2021 – 05/2022

- Developed features for DHL Freight's **Transport Management System (TMS)**, which runs customers' end-to-end transport operations across DHL's regional and national network.
- **Automated logistics workflows**, reducing manual data entry and human error in shipment scheduling.
- Built deep **freight-logistics domain knowledge** through DHL's domain training, applied directly to TMS work.

*Tech:* Java, Hibernate, internal WexVs Designer/Client frameworks

**Software Engineer** **Rubikal** Alexandria, Egypt 01/2021 – 06/2022

- **Redesigned the data pipeline**, cutting average ETL time by **45%** and enabling real-time analytics.
- Analyzed education data covering **millions of students** and delivered actionable insights to clients.
- Built **Tableau and Metabase dashboards** that presented findings clearly to client stakeholders.
- Applied the **GTFS** transit-feed standard so public transit agency Mowasalat Misr could publish its transit data.
- Built and deployed serverless functions on **AWS Lambda**.

*Tech:* Spring, Rails, Python, SQL, Snowflake, dbt, AWS, Metabase, Tableau, Jira

**Cloud Native Engineer (Part-time)** **Cassbana** Cairo, Egypt 10/2020 – 02/2021

- Led the migration of monolithic applications to a **serverless architecture**, improving scalability and cutting operational costs by **35%**.
- Rebuilt a legacy PHP/Laravel codebase as **Azure Functions** with Entity Framework, reducing latency by **40%** without breaking existing integrations.
- Automated **GitLab CI/CD** deployment pipelines, cutting deployment time by **60%** and reducing production errors.
- Optimized **Cosmos DB** data access, speeding up retrieval by **25%** while maintaining high availability.

*Tech:* Azure Functions, Entity Framework, Cosmos DB, Laravel, GitLab CI/CD, Serverless

**Software Engineer** **Sia** Cairo, Egypt 04/2020 – 12/2020

- Built **7+ microservices** (gamification, notifications, educational content) for a 10-person ed-tech startup.
- Created an **end-to-end CI/CD pipeline** that cut deployment time by **60%**, enabling much faster feature releases.
- Optimized database queries, improving data-retrieval times by **30%** and speeding up responses for users.

*Tech:* Kotlin, Spring Boot, Angular, AWS, Docker, Jenkins, Jira

**Software Engineer (Part-time)** **Robo-Tech** Alexandria, Egypt 08/2018 – 04/2020

- Built the communication system for **remotely operated vehicles (ROVs)**, reducing packet loss to below **0.01%**.
- Designed and launched **2 ROV prototypes** that passed testing in varied underwater conditions and won client approval for mass production.
- Improved real-time object-detection accuracy by **30%** by optimizing the vehicle's **OpenCV**-based navigation system.
- Cut prototype delivery time by **20%** by shortening development cycles with a cross-functional engineering team.

*Tech:* C++, Qt, C#, OpenCV

## Technical Skills

**Core Competencies**: System design, distributed systems, automated testing, CI/CD, observability, cloud-native development, LLM applications & AI agents, technical leadership, mentoring

**Programming Languages**: Java, Kotlin, Python, TypeScript, SQL, C++, Ruby

**Backend**: Spring Boot, Hibernate/JPA, Django, FastAPI, Node.js, Rails, Celery, JUnit, Testcontainers

**Data**: PostgreSQL, Redis, MongoDB, TimescaleDB, Snowflake, dbt, ETL pipelines, Metabase

**Architecture**: Microservices, event-driven architecture, Kafka (Nakadi), REST API design, OPA/Rego, multi-tenancy

**Cloud & Operations**: AWS (ECS, RDS, Lambda, S3, Textract), CloudFormation, Azure, Kubernetes, Docker, CI/CD, OpenTelemetry, Grafana

**AI & Agentic Engineering**: Claude Code, GitHub Copilot, MCP, multi-agent orchestration, OpenAI API, LangChain, RAG, Pinecone, embeddings, n8n

**Frontend & Mobile**: React, Next.js, React Native, Angular

**Spoken Languages**: Arabic (native), English (fluent), German (B1), French (A1)

## Education

**B.Sc. Computer & Communication Engineering, Alexandria University** Alexandria, Egypt

## Certifications

Understanding Agentic AI (Digital Workforce Services)

AWS Fundamentals: Building Serverless Applications (Amazon Web Services)

Getting Started with Google Kubernetes Engine (Google Cloud, Coursera)

Neural Networks and Deep Learning (Coursera)

Spring: Ecosystem and Core (LearnQuest)